4.2.1 Prove $E_k^{\bar{v}}(\tau) := \varepsilon_N \sum_{\substack{(c,d)\equiv v(N)\\ \gcd(c,d)=1}} (c\tau+d)^{-k} = \varepsilon_N \sum_{\gamma\in(P_+\cap\Gamma(N))\backslash\Gamma(N)\delta} j(\gamma,\tau)^{-k}$.

as representative of coset.

Pf: Only need to show

$$\left.\begin{matrix}(c,d)\equiv v\ (N)\\ \gcd(c,d)=1\end{matrix}\right\} \iff \gamma\in(P_+\cap\Gamma(N))\backslash\Gamma(N)\delta.$$

$\gamma = \begin{pmatrix} a' & b' \\ c & d \end{pmatrix}$.

Recall that $P_+ = \left\{\begin{pmatrix}1 & n\\ 0 & 1\end{pmatrix} : n\in\mathbb{Z}\right\}$. $\delta = \begin{pmatrix} a & b\\ c_v & d_v\end{pmatrix}\in SL_2(\mathbb{Z})$. $(c_v,d_v)\equiv \bar{v}(N)$. $\equiv v$

then $P_+\cap\Gamma(N) = \left\{\begin{pmatrix}1 & kN\\ 0 & 1\end{pmatrix} : k\in\mathbb{Z}\right\}$.

$$\Gamma(N)\delta = \left\{\cancel{\begin{pmatrix} k_1Na & k_2Nb\\ k_3Nc_v & k_4Nd_v\end{pmatrix} : k_i\in\mathbb{Z}}\ X\equiv\begin{pmatrix} a & b\\ c_v & d_v\end{pmatrix}(N)\right\}$$

So $(P_+\cap\Gamma(N))\backslash\Gamma(N)\delta = \left\{(P_+\cap\Gamma(N))X : X\equiv\begin{pmatrix} a & b\\ c_v & d_v\end{pmatrix}(N)\right\}$

Thus $\forall\gamma\in(P_+\cap\Gamma(N))\backslash\Gamma(N)\delta$, $\gamma = \begin{pmatrix} a+kN\cdots & b+N\cdots\\ c_v+N\cdots & d_v+N\cdots\end{pmatrix}$.

a representative could be $\begin{pmatrix} a & b\\ c & d\end{pmatrix}$ with $(c,d)\equiv v\ (N)$, $\gcd(c,d)=1$

which is of simplest form.

4.2.2 Let $N, \bar{v}, k$. Let P be of level N. Define

$$E_{k,P}^{\bar{v}} := \sum_{\gamma_j\in\Gamma(N)\backslash P} E_k^{\bar{v}}[\gamma_j]_k,$$

Sum over coset representatives. Show well defined and $E_{k,P}^{\bar{v}}\in M_k(P)$.

Pf: If $\gamma_j, \delta_j\in\Gamma(N)\backslash P$ then $\exists\ \varepsilon_j\in\Gamma(N)$ $\delta_j = \varepsilon_j\gamma_j$.

$$\bigcup_j \Gamma(N)\gamma_j = \bigcup_j\Gamma(N)\delta_j = P.$$

$$\Rightarrow E_k^{\bar{v}}[\delta_j]_k = E_k^{\bar{v}}[\varepsilon_j]_k[\gamma_j]_k \overset{\text{prop 4.2.1}}{=} E_k^{\overline{v\varepsilon_j}}[\gamma_j]_k \quad \forall j.$$

$$= \varepsilon_N \sum_{\gamma\in(P_+\cap\Gamma(N))\backslash\delta\Gamma(N)\varepsilon_j} j(\gamma,\tau)^{-k} = \varepsilon_N\sum_{\gamma\in(P_+\cap\Gamma(N))\backslash\Gamma(N)\delta} j(\gamma,\tau)^{-k} = E_k^{\bar{v}}[\gamma_j]_k.$$

We want to show $\sum_{\gamma_j\in\Gamma(N)\backslash P} E_k^{\bar{v}}[\gamma_j]_k = \sum_{\gamma_j\in\Gamma(N)\backslash P} E_k^{\bar{v}}[\gamma_j]_k[\gamma]_k \quad \forall\gamma\in P$.

$$\bigcup_j \Gamma(N)\gamma_j = P \Rightarrow \bigcup_j \Gamma(N)(\gamma_j\gamma) = \left(\bigcup_j\Gamma(N)\gamma_j\right)\gamma = P\gamma = P.$$

(4.2.3) Establish formula in the section for $\lim_{\mathrm{Im}\,\tau\to\infty} E_k^{\bar v}(\tau)$:

$$\lim_{\mathrm{Im}(\tau)\to\infty} E_k^{\bar v}(\tau) = \begin{cases} (\pm 1)^k & \bar v = \pm\overline{(0,1)}, \text{ unless } k \text{ odd and } N=1,2. \\ 0 & \text{otherwise.} \end{cases}$$

Pf: If $\bar v = \pm\overline{(0,1)}$, then

$$E_k^{\bar v}(\tau) = \epsilon_N \sum_{\substack{(c,d)\equiv\pm(0,1)\,(N) \\ \gcd(c,d)=1}} \frac{1}{(c\tau+d)^k} = (\pm 1)^k + \sum_{\substack{(c,d)\equiv\pm(0,1)\,(N) \\ \gcd(c,d)=1 \\ (c,d)\neq\pm(0,1)}} \frac{1}{(c\tau+d)^k}$$

$$\to (\pm 1)^k.$$

otherwise every term goes to 0.

If $N=1$, then $E_k^{\bar v}(\tau) = \frac{1}{2}\sum'_{\substack{c,d\in\mathbb{Z} \\ \gcd(c,d)=1}} \frac{1}{(c\tau+d)^k} \to \frac{1}{2}\sum_{d\in\mathbb{Z}-\{0\}} \frac{1}{d^k}$

If $N=2$, then $E_k^{\bar v}(\tau) = \frac{1}{2}\sum_{\substack{(c,d)\equiv\pm(0,1)\,(2) \\ \gcd(c,d)=1}} \frac{1}{(c\tau+d)^k} \xrightarrow{d=2t+1} \frac{1}{2}\sum_{t\in\mathbb{Z}} \frac{1}{(2t+1)^k}$.

(4.2.4) (a) Möbius function $\mu: \mathbb{Z}_+^* \to \mathbb{Z}$

$$n \mapsto \begin{cases} (-1)^g & n = p_1\cdots p_g \text{ distinct primes} \\ 0 & p^2 \mid n \quad \exists p \text{ prime}. \end{cases}$$

with $\mu(1)=1$.

Show $\forall n\in\mathbb{Z}_+$ $\sum_{\substack{d\mid n \\ d>0}} \mu(d) = \begin{cases} 1 & n=1 \\ 0 & n>1. \end{cases}$ (Möbius inversion)

Pf: This is to shown for by for $n = p_1,\dots,p_g \neq 1$ distinct,

$$1 - \binom{g}{1} + \binom{g}{2} - \cdots + (-1)^g\binom{g}{g} = 0.$$

For $p^2 \mid n \neq 1$, pick all such $p_1,\dots,p_t$, let n' ~~=~~

with $p_i^{r_i} \| n$, let $n' = \frac{n}{p_1^{r_1}\cdots p_t^{r_t}}$.

Then $\sum_{\substack{d\mid n \\ d>0}} \mu(d) = \sum_{\substack{d\mid n' \\ d>0}} \mu(d) = 0.$

(4.2.4) (b) Let $\bar{v} \in (\mathbb{Z}/N\mathbb{Z})^2$ be point of order N.

From (a), $E_k^{\bar{v}}(\tau) = \epsilon_N \sum' \sum_{\substack{(c,d)\equiv v\ (N) \\ n \mid \gcd(c,d) \\ n>0}} \frac{\mu(n)}{(c\tau+d)^k}$.

Rearrange the sum to obtain formula (4.6):

$$E_k^{\bar{v}}(\tau) = \epsilon_N \sum_{n\in(\mathbb{Z}/N\mathbb{Z})^\times} \zeta_+^n(k,\mu)\, G_k^{n^{-1}\bar{v}}(\tau).$$

where $\zeta_+^n(k,\mu) = \sum_{\substack{m=1 \\ m\equiv n\ (N)}}^{\infty} \frac{\mu(m)}{m^k}$ $\quad n\in(\mathbb{Z}/N\mathbb{Z})^\times$.

Pf: Since $\bar{v}$ is of order N, $\gcd(\gcd(c_v,d_v),N)=1$.

$\Rightarrow n:=\gcd(c,d) \in (\mathbb{Z}/N\mathbb{Z})^\times$. $\quad m \mid \gcd(c,d)$ $\Rightarrow m\equiv n\ (N)$.

$$E_k^{\bar{v}}(\tau) = \epsilon_N \sum_{n>0} \mu(n) \sum_{\substack{(c,d)\equiv \bar{v}\ (N) \\ n\mid \gcd(c,d)}} \frac{1}{(c\tau+d)^k}$$

$$= \epsilon_N \sum_{n>0} \frac{\mu(n)}{n^k} \sum_{\substack{(c',d') \\ (nc',nd')\equiv \bar{v}(N)}} \frac{1}{(c'\tau+d')^k}$$

$$= \epsilon_N \sum_{n>0} \frac{\mu(n)}{n^k} \sum_{(c',d')\equiv n^{-1}\bar{v}(N)} \frac{1}{(c'\tau+d')^k} = \epsilon_N \sum_{\substack{n>0 \\ (n,N)=1}} \frac{\mu(n)}{n^k} G_k^{n^{-1}\bar{v}}(\tau)$$

$$= \epsilon_N \sum_{a\in(\mathbb{Z}/N\mathbb{Z})^\times} \zeta_+^a(k,\mu)\, G_k^{a^{-1}\bar{v}}(\tau).$$

4.2.5 (a) Show constant term of (4.7) $G_k^{\bar{v}}(\tau) = \sum \sum_{d\in\mathbb{Z}} \frac{1}{(c\tau+d_v+Nd)^k}$

is given by (4.8) $\begin{cases} 1 & \bar{c}_v = \bar{0} \equiv c_v \\ 0 & \text{otherwise} \end{cases} \cdot \sum'_{d\equiv d_v} \frac{1}{d^k}$.

Sol: $\bar{c}_v = \bar{0} \Rightarrow$ const $= \sum_{d\in\mathbb{Z}} \frac{1}{(d_v+Nd)^k} = \sum'_{d\equiv d_v\ (N)} \frac{1}{d^k}$

$\bar{c}_v \neq \bar{0} \Rightarrow$ ~~const~~ all terms has $\frac{1}{(c\tau+\cdots)^k}$ τ inside

const $=0$.